

Liar's Deck

Project overview (Marp)

ASE 285 — Individual project slides (hosted HTML)

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Problem

Multiplayer games need **trusted rules** and **private cards**. Client-only logic invites cheating and leaks.

Solution

- **Socket.IO** real-time sync
- **Server-authoritative** dealing, turns, and Call Liar
- **Partial state** to each client + session tokens

Stack

React + Vite · Express + Socket.IO · MongoDB (optional persistence)

Project metrics

Scope & completion

Metric	Planned	Delivered
Features	6	6
Requirements (R1.1–R6.3)	18	18

Testing

Backend suites under `src/backend/tests/` — `gameLogic.test.js`, `socket.test.js`, `regression.test.js`

Requirements trace to acceptance scenarios in `docs/requirements/`

Sprint burndown

Sprint 1

Metric	Planned	Completed	Burndown
Features	3	3	100%
Requirements	9	9	100%

Delivered: F1 Lobby · F2 Dealing · F3 Turn-based play

Sprint 2

Metric	Planned	Completed	Burndown
Features	3	3	100%
Requirements	9	9	100%

Delivered: F4 Bluff detection · F5 Hidden state / reconnect · F6 Persistence

Lines of code

Primary sources (line counts)

Counts are primary application sources; totals drift as the codebase evolves.

As of April 2026.

File	Lines	Role
src/backend/app.ts	522	Express, Socket.IO, rooms, Call Liar, sessions
src/backend/gameLogic.ts	188	Shuffle, deal, hide/show hands, rounds, roulette
src/backend/mongoPersistence.ts	76	MongoDB load / persist room snapshots
src/backend/index.ts	18	Server entry, graceful Mongo shutdown
src/frontend/src/App.tsx	700	React UI, lobby, game, Call Liar
src/frontend/src/App.css	1243	Layout and card styling
src/frontend/src/socket.ts	9	Socket.IO client
Total	2756	

Features delivered (F1–F6)

ID	Feature
F1	Matchmaking and lobby system
F2	Server-authoritative card dealing
F3	Turn-based gameplay and validation
F4	Bluff detection (Call BS / Call Liar)
F5	Hidden state and partial state views
F6	Session persistence and reconnection

Sprint highlights

Sprint 1

- Scaffolding — Express, React + Vite, Socket.IO
- Lobby (join, list, ready) and real-time events
- Dealing, hidden hands, per-player views
- Turn order, play UI, validation

Sprint 2

- Call Liar flow, truth check, penalties / rounds
- Reconnect privacy (`tryRestoreSession` , hidden broadcasts)
- MongoDB persistence and snapshot load
- Graceful disconnect, regression tests, polish

Links & artifacts

Resource	Location
GitHub repository	<code>https://github.com/joseph-ampfer/Bluff_Multiplayer</code>
Requirements	<code>docs/requirements/requirements.md</code>
Sprint 1 review (Marp)	<code>docs/presentation/sprint1_review.md</code>
Sprint 2 review (Marp)	<code>docs/presentation/sprint2_review.md</code>
GitHub Pages	Site root / docs index (see repo workflow)

Thank you

Liar's Deck — real-time multiplayer bluffing card game